University of Applied Sciences

HOCHSCHULE EMDEN·LEER

Data Preparation and Variable Selection

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Agenda

1. Introduction
2. Data preliminary treatment
3. Variable assessment
4. Data second treatment
5. Multi-variable selection
6. Summary

Introduction

Introduction

A working process for statistic research

- After collecting the data
- Before further researching for more specific target

A working process for statistic research

Preliminary treatment

- Scaling method
- Binning method

→

Variable assessment

- Weight of evidence
- Information value

→

Second treatment

- Identify unreasonable data
- Missing data handling
- Outlier handling

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Multi-variable selection

- Selection strategy
- Criteria choosing
- Setting the benchmark

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Why we need preliminary process?

The meaning behind initial treatment: the „first investigation“

- Any strange data occur? (i.e. outlier)
- The major feature of our data distribution
- Researcher prefer „normal (or standardized) distribution“

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Scaling methods

Here, we present three scalings:

- Z scaling
- Min-Max scaling
- Lognormal scaling

Z scaling

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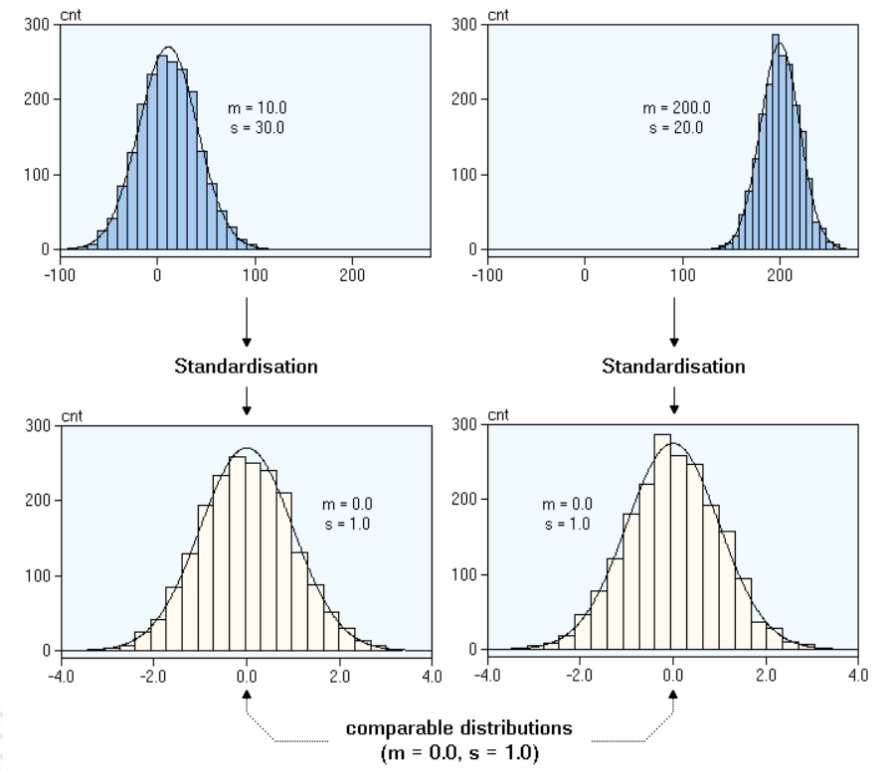

Z-scaling example

Min-Max scaling

$$x_i' = \frac{x_i - x_{min}}{x_{max} - x_{min}} \tag{2}$$

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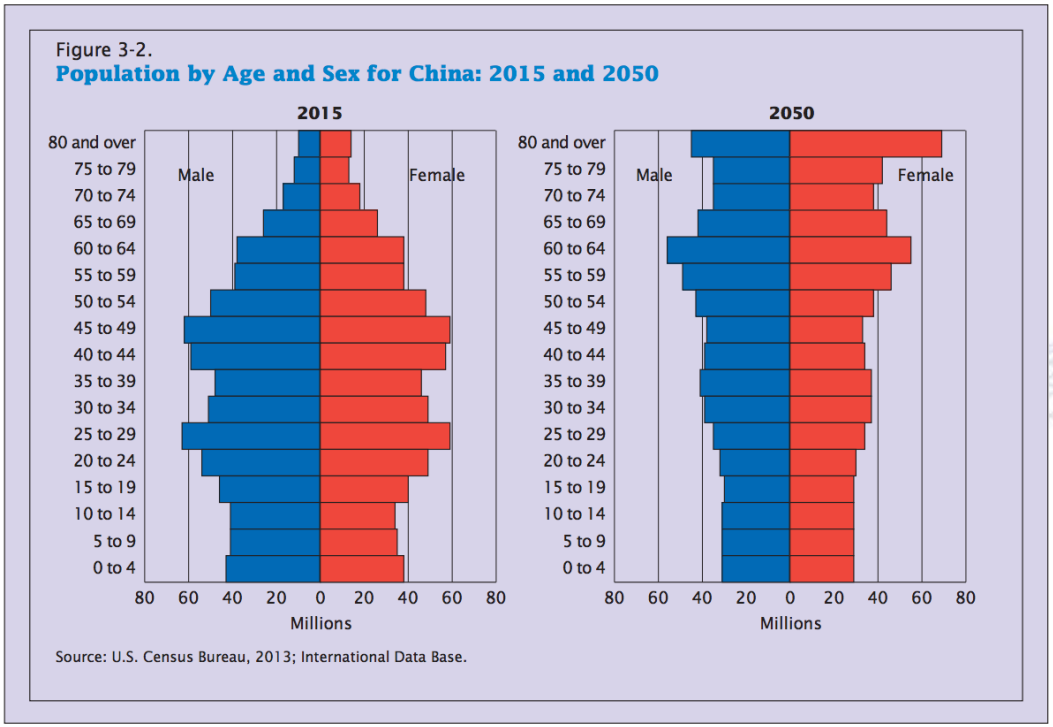

Min-Max scaling example

Lognormal scaling

$$x_i' = \ln(x_i) \tag{3}$$

A special scaling trick which can transform the distribution curve into more bell-curve-like (more similar to normality) distribution.

The comparison of scaling methods

Variable scaling method	Equation	Suitable case
Z scaling	$x_i' = \frac{x_i - \overline{x}}{\sigma}$	- Variable variance study - Normal distribution case - similar to normality
Min-Max scaling	$x_i' = \frac{x_i - x_{min}}{x_{max} - x_{min}}$	- Scale of variable is less important - Two data set comparison with same scale of variable
Lognormal scaling	$x_i' = \ln(x_i)$	- The distribution type are logarithm distribution or similar to it

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Initial evaluation of variable

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- A burden workload during late research period
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- A burden workload during late research period
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A tool for „pre-eliminate“ the amount of variable

$\longrightarrow$ To find out if this variable relevant enough in terms of correlation between the target variable (the object to be predicted)

Example: production defect issue

Machine No.	Sampling Qty	Qty of NG	Qty of Good
#1	22	7	15
#2	13	4	9
#3	16	5	11
#4	17	5	12
#5	38	12	26
#6	11	3	8
#7	20	6	14
#8	25	7	18
#9	23	7	16
#10	15	4	11
Total	200	60	140

Machine case 1(Data)

Example: production defect issue

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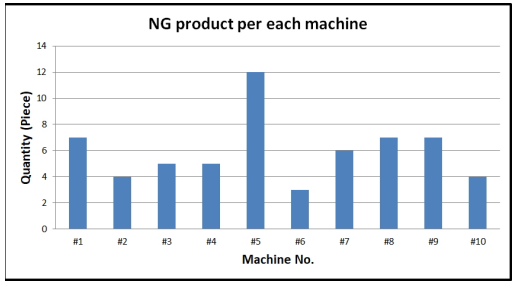

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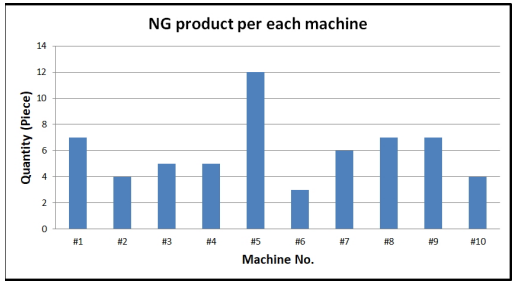

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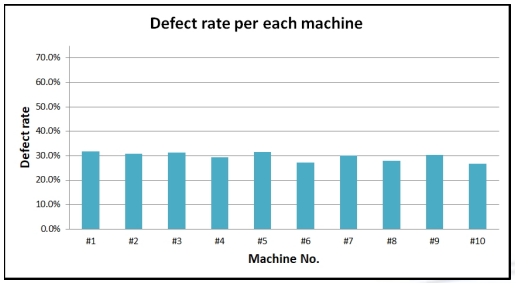

Machine case 1(Defect rate)

Weight of Evidence (WoE)

$$WoE = \ln\left(\frac{dist(G)}{dist(B)}\right) \tag{4}$$

If b ist the number of bads concerning the parameter, b_T the total number of bads, then we get

$$dist(B) := \frac{b}{b_T}$$

If g ist the number of goods concerning the parameter, g_T the total number of goods, then we get

$$dist(G) := \frac{g}{g_T}$$

Weight of Evidence (WoE) - Thumb Rule

$$WoE = \ln\left(\frac{dist(G)}{dist(B)}\right)$$

Information Value	**Predictive Power**
< 0.02	useless for prediction
0.02 to 0.1	Weak predictor
0.1 to 0.3	Medium predictor
0.3 to 0.5	Strong predictor
> 0.5	Suspicious or too good to be true

Typically, variables with medium and strong predictive powers are selected for model development.

Weight of Evidence (WoE) - Example

Machine No.	Sampling Qty	Qty of Bad	Qty of Good	Bad Rate	Dist(B)	Dist(G)	WoE
#1	22	7	15	31,8%	0,117	0,107	-0,085
#2	13	4	9	30,8%	0,067	0,064	-0,036
#3	16	5	11	31,3%	0,083	0,079	-0,059
#4	17	5	12	29,4%	0,083	0,086	0,028
#5	38	12	26	31,6%	0,200	0,186	-0,074
#6	11	3	8	27,3%	0,050	0,057	0,134
#7	20	6	14	30,0%	0,100	0,100	0,000
#8	25	7	18	28,0%	0,117	0,129	0,097
#9	23	7	16	30,4%	0,117	0,114	-0,021
#10	15	4	11	26,7%	0,067	0,079	0,164
Total	200	60	140	29,7%			

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Total	200	60	140	29,7%			

The „contribution“ about how much deviation each machine has comparing to the overall defect rate

Information value (IV) for Intervals

$$IV = \sum_i \left[(dist(G)_i - dist(B)_i) \cdot \ln \left(\frac{dist(G_i)}{Dist(B_i)} \right) \right] \tag{5}$$

- Integrate the WoE from each interval of one variable
- Quantify the influence of a variable on target variable

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Data second treatment

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- Scaling method
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Variable assessment
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- Identify unreasonable data
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Multi-variable selection
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The nature flaw of data collection

Why unreasonable or insufficient data occur:

- The false data due to technical issue (i.e. the limit of measurement device)
- Interview providing wrong information (accidently or intentional)
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$\longrightarrow$ **Missing data and outlier issue**

The action:
1. Identify the pattern of flaw
2. Then deal with it accordingly

Example: Asthma survey

154 interviewees, 7 questions(variables)
$\longrightarrow$ Theoretically 154$\times$7 data value available

Variable	Definition	Possible values	*M*	(*SD*)	*N*
Asthma belief Survey	Level of confidence in controlling asthma	Range from 1, little confidence to 5, lots of confidence	4.057	(0.713)	154
Group	Treatment or control group	0 = Treatment 1 = Control	0.558	(0.498)	154
Symsev	Severity of asthma symptoms in 2 week period post-treatment	0 = no symptoms 1 = mild symptoms 2 = moderate symptoms 3 = severe symptoms	0.235	(0.370)	141
Reading	Standardized state reading test score	Grade equivalent scores, ranging from 1.10 to 8.10	3.443	(1.636)	79
Age	Age of child in years	Range from 8 to 14	10.586	(1.605)	152
Gender	Gender of child	0 = Male 1 = Female	0.442	(0.498)	154
Allergy	Number of allergies reported	Range from 0 to 7	2.783	(1.919)	83

Asthma study

Symsev	Reading	Age	Allergy	# of cases	% of cases
O	O	O	O	19	12.3
M	O	O	O	1	0.6
O	M	O	O	54	35.1
O	O	O	M	56	36.4
M	M	O	O	9	5.8
M	O	O	M	1	0.6
O	M	O	M	10	6.5
O	O	M	M	2	1.3
M	M	O	M	2	1.3
# missing 13 (8.4%)	# missing 75 (48.7%)	# missing 2 (1.3%)	# missing 71 (46.1)	154	

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Asthma study(Missing pattern)

Only 19 people answer all the questions!

Treatment of missing data

- **Complete case analysis:**
  only adopts the case who provide the required data to all variables.
- **Available case analysis:**
  counts every recorded data from all the case.
- **Imputation:**
  Make a „virtual data“ to substitute the missing one.

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- **Trimming:**
  Not consider the outlier.
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  (i.e. put a weight on mean value)

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Aim at focusing critical predictor variable

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To make the regression model as simple as possible, only keep those variables which is independent enough and being able to demonstrate the correlation between itself and the target variable.

1. Make the model **"lean"**
2. Perform **best fitting** to reality

The selection strategy

Step 1. Decide the selection approach (i.e. forward, backward, stepwise etc.)

Step 2. Choose the criteria for selection (i.e. Standard deviation, MSE, Z-score, probability etc.)

Step 3. Determine which index is the „judge tool“ and set a benchmark as stop point of selection

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Index introduction: Z-Score / T-score -1

$$Z = \frac{\overline{X} - \mu_0}{\sigma / \sqrt{n}} \tag{6}$$

- $\overline{X}$: mean value of a predictor variable
- μ_0: mean value of model's population.
- σ: represent the standard deviation of model's population
- n: the sample number of a variable

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It means how „independent“ our predictor variable is in terms of the correlation between the predictor and the model data set.

Index introduction: Z-Score / T-score -2

Z-score is not valid for two condition:

- Unknown standard deviation of model's population
- Smaller quantity of sample data (≤ 30)

Use „T-score“ as alternative:

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Z-score is not valid for two condition:

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Use „T-score“ as alternative:

$$T = \frac{\overline{X} - \mu_0}{\sigma_s / \sqrt{n}} \tag{7}$$

$\longrightarrow$ σ_s: **Standard deviation of samples**

Index introduction: Mean square error (MSE)

$$MSE(x) = \frac{1}{n}\sum_{i=1}^{n}(X_{sim} - x_i)^2 \tag{8}$$

- x_i: the real value of observed data
- X_{sim}: the theoretical simulated data base on modeling
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- X_{sim}: the theoretical simulated data base on modeling
- n: the sample number of all variable

It means how much extent those observed data deviate away from the correspondent simulated value.

Selection approach

1. **Forward selection:**
   - Start the selection with a model which has **"no variable" (empty model)**
   - **Add** only one variable per each round base on criteria
   - Stop adding variable when the whole model reach the benchmark
2. **Backward selection:**
   - Start the selection with a model which is **"full of all variables"**
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Backforward selection example

A linear regression study:
5 type of personality test as predictor variable, IQ test as target variable.

- Criteria: T-score (prefer to away from zero)
- Benchmark: Square of MSE (set maximum limit 10)

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Backforward selection example - 1st round

In	Variable	Standard. Coefficient	R-Squared Increment	R-Squared Other X's	T-Value	Prob Level	Pct Change Sqrt(MSE)
Yes	Test1	-3.0524	.2357	.9747	-1.8789	.0930	11.9387
Yes	Test2	-2.9224	.2414	.9717	-1.9016	.0897	12.3210
Yes	Test3	.1404	.0152	.2280	.4773	.6445	-3.9386
Yes	Test4	4.7853	.2832	.9876	2.0596	.0695	15.0741
Yes	Test5	-.0595	.0027	.2329	-.2017	.8447	-4.9176
R-Squared=0.3991		Sqrt(MSE)=10.65198					

Backforward selection example - 1st round

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Yes	Test2	-2.9224	.2414	.9717	-1.9016	.0897	12.3210
Yes	Test3	.1404	.0152	.2280	.4773	.6445	-3.9386
Yes	Test4	4.7853	.2832	.9876	2.0596	.0695	15.0741
Yes	Test5	-.0595	.0027	.2329	-.2017	.8447	-4.9176
R-Squared=0.3991		Sqrt(MSE)=10.65198					

→ **rule out test 5 (T= -0.2017)**

Backforward selection example - 2nd round

In	Variable	Standard. Coefficient	R-Squared Increment	R-Squared Other X's	T-Value	Prob Level	Pct Change Sqrt(MSE)
Yes	Test1	-3.0612	.2373	.9747	-1.9825	.0756	12.5340
Yes	Test2	-2.9032	.2392	.9716	-1.9906	.0745	12.6640
Yes	Test3	.1163	.0125	.0752	.4550	.6588	-3.6717
Yes	Test4	4.7850	.2832	.9876	2.1660	.0555	15.5681
No	Test5		.0027	.2329	.2017	.8447	5.1719
R-Squared=0.3964		Sqrt(MSE)=10.12816					

Backforward selection example - 2nd round

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Yes	Test4	4.7850	.2832	.9876	2.1660	.0555	15.5681
No	Test5		.0027	.2329	.2017	.8447	5.1719
R-Squared=0.3964		Sqrt(MSE)=10.12816					

$\rightarrow$ **rule out test 3 (T= 0.455)**

Backforward selection example - 3rd round

In	Variable	Standard. Coefficient	R-Squared Increment	R-Squared Other X's	T-Value	Prob Level	Pct Change Sqrt(MSE)
Yes	Test1	-3.1020	.2444	.9746	-2.0890	.0607	13.1519
Yes	Test2	-2.9024	.2391	.9716	-2.0659	.0632	12.7977
Yes	Test4	4.7988	.2849	.9876	2.2553	.0455	15.7808
No	Test3		.0125	.0752	.4550	.6588	3.8116
No	Test5		.0000	.0810	.0087	.9932	4.8805
R-Squared=.3839		Sqrt(MSE)=9.756291					

Backforward selection example - 3rd round

In	Variable	Standard. Coefficient	R-Squared Increment	R-Squared Other X's	T-Value	Prob Level	Pct Change Sqrt(MSE)
Yes	Test1	-3.1020	.2444	.9746	-2.0890	.0607	13.1519
Yes	Test2	-2.9024	.2391	.9716	-2.0659	.0632	12.7977
Yes	Test4	4.7988	.2849	.9876	2.2553	.0455	15.7808
No	Test3		.0125	.0752	.4550	.6588	3.8116
No	Test5		.0000	.0810	.0087	.9932	4.8805
R-Squared=.3839		Sqrt(MSE)=9.756291					

→ **Sqrt(MSE) reach the benchmark! (less than 10)**
→ **Finalize the number of 3 variables**

Summary

Summary and learning experience

- The working procedure of statistic research is flexible but highly rely on related professional knowledge of certain area.
- The idea is similar to production management: prevent deficient data/variable (raw material/ work in-process) from early stage of process